

MODULE 1

Answer any TWO questions.

1. (a) (i) Name a non-native species that was introduced in a named country. [2 marks]
- (ii) State THREE characteristics of a successful non-native species. [3 marks]
- (b) Describe THREE impacts of the introduction of a non-native species into an ecosystem. [9 marks]
- (c) Suggest THREE mitigating measures that can be used to deal with introduced species. [6 marks]

Total 20 marks

2. (a) Figure 1 below is based on energy flow in a cornfield in a field lot. It shows the amount of energy that is present in a trophic level in various feeders.

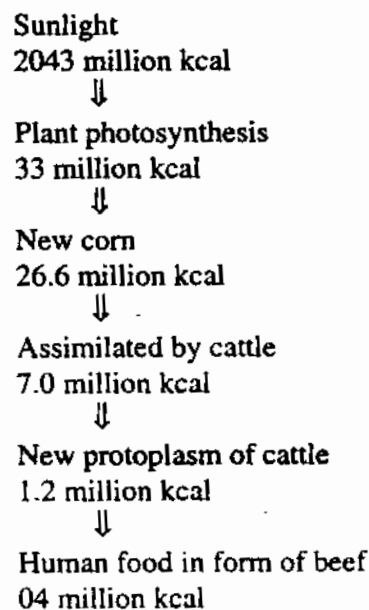


Figure 1: Energy flow in a cornfield in a field lot
(Based on Transeau 1926 and Heichel 1976)

GO ON TO THE NEXT PAGE

(i) Name THREE processes through which energy is lost at EACH level.

[3 marks]

(ii) Assess whether or not it is more energy-efficient for humans to feed on corn than on cattle.

[7 marks]

(b) (i) Explain the term 'productivity of producers'.

[6 marks]

(ii) Discuss TWO implications of inefficiency of energy transfer through a food chain.

[4 marks]

Total 20 marks

3. (a) (i) Define the term 'ecosystem'.

[3 marks]

(ii) State TWO major processes occurring within an ecosystem.

[2 marks]

(b) Describe THREE abiotic factors that have a direct impact on organisms in a chosen aquatic ecosystem.

[15 marks]

Total 20 marks

GO ON TO THE NEXT PAGE

MODULE 2

Answer any TWO questions.

4. (a) Discuss the relationship between population growth and poverty. [10 marks]
- (b) Describe TWO factors affecting human population growth rate. [10 marks]

Total 20 marks

5. (a) Explain why the consumption patterns of human population is a root cause of environmental degradation. [14 marks]
- (b) Describe the current geographical variation in the consumption patterns of human population. [6 marks]

Total 20 marks

6. (a) Explain what is meant by the term 'sustainable development'. [6 marks]
- (b) Identify and explain FOUR changes that may be implemented in a country to bring about sustainable development. [8 marks]
- (c) Identify TWO fundamental factors that could hamper efforts to bring about sustainable development in a named country. Explain the effect of EACH of the factors identified. [6 marks]

Total 20 marks

GO ON TO THE NEXT PAGE

MODULE 3

Answer any TWO questions.

7. Describe how EACH of the factors below affect natural resource use in the Caribbean.

- (a) Demographic [4 marks]
- (b) Geographic [4 marks]
- (c) Political [6 marks]
- (d) Economic [6 marks]

Total 20 marks

8. Write an essay (about 400 words) on the environmental impacts resulting from the use of EITHER minerals OR lands and beaches as natural resources of the Caribbean, citing specific examples.

Total 20 marks

9. In about 400 words, justify the need for natural resource conservation in your country.

Total 20 marks

END OF TEST

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